MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – CHAPTER TEST (SET – 9)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 7: Coordinate Geometry** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The distance of the point $P(-6, 8)$ from the origin is: [1]

(A) 8 (B) $2\sqrt{7}$ (C) 10 (D) 6

Q2. The distance of which of the following points from origin is less than 5 units ? (a) $(3, 4)$ (b) $(2, 6)$ (c) $(-3, -4)$ (d) $(1, 4)$ [1]

Q3. The distance of point $P(1, -1)$ from the x-axis is: [1]

(A) 1 (B) -1 (C) 0 (D) $\sqrt{2}$

Q4. The point on the x-axis which is equidistant from the points $A(-1, 0)$ and $B(5, 0)$ is: [1]

(A) $(2, 0)$ (B) $(0, 2)$ (C) $(3, 0)$ (D) $(-2, 0)$

Q5. Find the distance between the points $A(0, 6)$ and $B(0, -2)$. [1]

(A) 6 units (B) 8 units (C) 4 units (D) 10 units

SECTION B: Short Answer Type Questions (6 Marks)

Q6. Find the length of the median through the vertex B of ΔABC with vertices $A(9, -2)$, $B(-3, 7)$ and $C(-1, 10)$. [2]

Q7. If (a, b) is the mid-point of the line segment joining the points $A(10, -6)$ and $B(k, 4)$ and $a - 2b = 18$, then find the value of k. [2]

Q8. Diagonals AC and BD of square ABCD intersect at P. Coordinates of points B and D are $(9, -2)$ and $(1, 6)$ respectively. (i) Find the co-ordinates of point P. (ii) Find the length of the side of the square. [2]

SECTION C: Long Answer & Competency Based Questions (9 Marks)

Q9. Vertices of a right triangle ABC with $\angle B = 90°$ are $A(3,4)$, $B(1,1)$ and $C(-8,7)$. Find the value of $\tan A$. ¿ **OR** ¿ (b) Using distance formula, prove that the points $A(2,3)$, $B(-7,0)$ and $C(-1,2)$ are collinear. ¿ ¿ [5]

Q10. Competency Case Study: Observe the map of Jaipur city placed on a Cartesian plane. Taking Rambagh Palace as origin, the location of some places are given below : Point A : (-4, 2) Rajasthan High Court Point B : (4, -4) Birla Mandir Point C : (4, 3) Heera Bagh Point D : (-5, -2) Amar Jawan Jyoti [4]

(i) Advocate Rehana stays at Heera Bagh. How much distance she has to cover daily to go to the court and coming back home ? [1 mark]

(ii) There is a crossing on X-axis which divides AD in a certain ratio. Find the ratio. [1 mark]

(iii) (a) Is Birla Mandir equidistant from Heera Bagh and Amar Jawan Jyoti ? Justify your answer. [2 marks] — **OR** —

(iv) (b) Using section formula, show that points A, O and B are not collinear. [2 marks]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★